

# CS5487 Machine Learning: Principles and Practice

Sem A 2025/26

Assitant Professor  
Kai Wang

Acknowledge to the course developer  
Prof. Antoni Chan



Department of  
Computer Science

香港城市大學  
City University of Hong Kong

## Prof. Antoni B. Chan

Professor, Dept. of Computer Science

Associate Dean (Research & Postgraduate), College of Computing

Deputy Director, Multimedia Engineering Research Centre (MERC)

BSc MEng Cornell, PhD UC San Diego

SrMIEEE

Precaution: 90% math + 10% coding

Machine Learning +  
Computer Vision + LeetCode

Kernel PCA

Data:  $\{x_1, \dots, x_n\}$

Feature:  $x_i \rightarrow \phi(x_i) = \phi_i$

Assume that data is centered in highdim f.s:  $\sum_{i=1}^n \phi(x_i) = 0$

1) Covariance matrix:

$$C = \frac{1}{n} \sum_{i=1}^n \phi(x_i) \phi(x_i)^T$$

2) Find eigen values / vectors  $(\lambda_j, v_j)$ :

$$C v_j = \lambda_j v_j$$

$$\frac{1}{n} \sum_{i=1}^n \underbrace{\phi(x_i) \phi(x_i)^T}_{\text{scalar } a_{ij}} v_j = \lambda_j v_j$$

Thus,  $v_j$  has the form:  $v_j = \sum_{i=1}^n a_{ij} \phi(x_i)$  (linear combo of  $\phi(x_i)$ )

$$v_j = \Phi a_j, \quad a_j = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{bmatrix}$$

# Instructors

- **Course Instructors**
  - Assistant Professor *Kai Wang*
    - Email: [kai.wang@cityu-dg.edu.cn](mailto:kai.wang@cityu-dg.edu.cn)
    - Phone: 15631906131
    - Office: AC4-505
    - Office Hours: Wednesday 9-11am

Who select both CS5487 and CS5491?

- CS5487: 90% math
- CS5491: 70% coding

# Lecture & Tutorial

- There is a three hour block for this course
  - The actual schedule will be tentative and marginally different from the official schedule.
  - *Lecture time will be 14:00-15:50.*
  - *Tutorial time will be 16:00-16:50*

|                        | Official | Actual          |
|------------------------|----------|-----------------|
| Tue<br>13:00-<br>14:00 | Lecture  | <b>Lecture</b>  |
| Tue<br>14:00-<br>15:00 | Lecture  | <b>Lecture</b>  |
| Tue<br>15:00-<br>15:50 | Tutorial | <b>Tutorial</b> |

# Lecture & Tutorial Format

## Lecture (face-to-face)

- Held in AC5-416

## Tutorial (face-to-face)

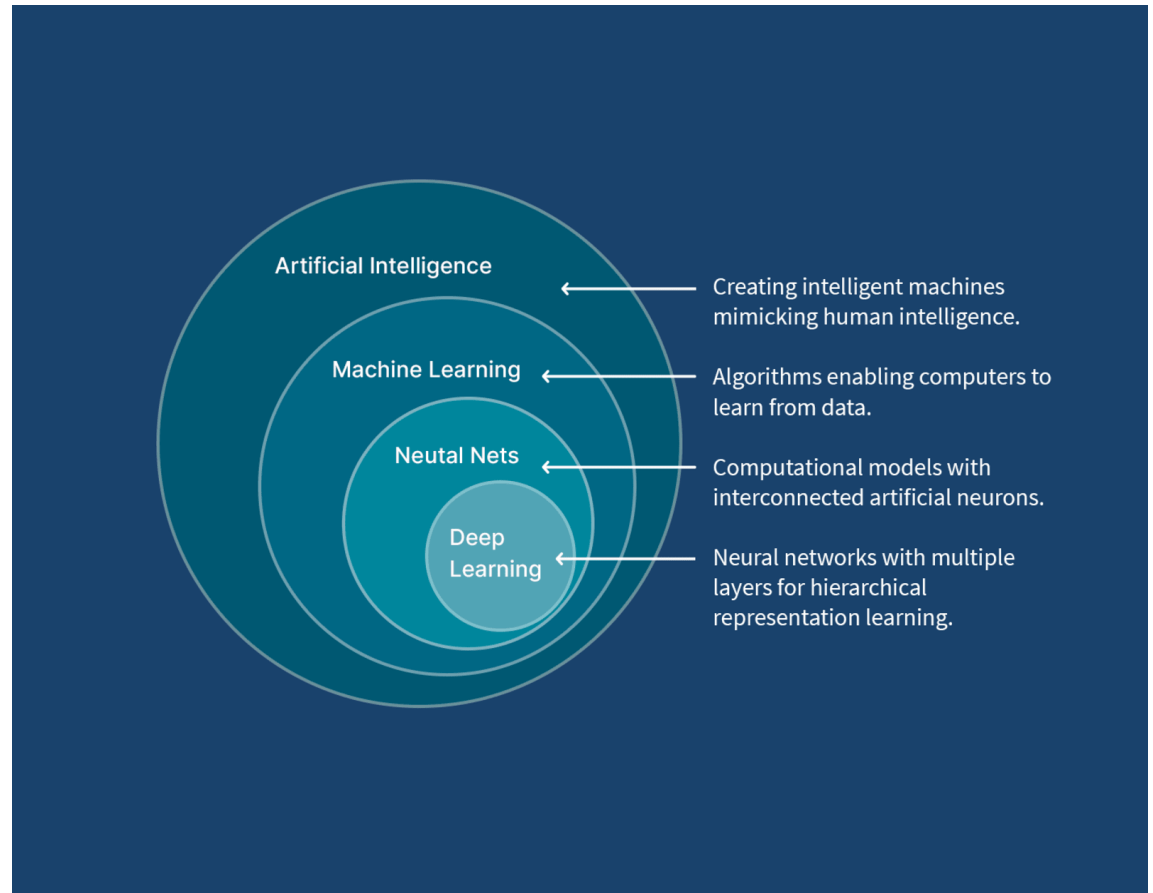
- Held in AC5-416

## Recording lectures/tutorials

- I will do my best to record the lectures and tutorials for student's to review later.

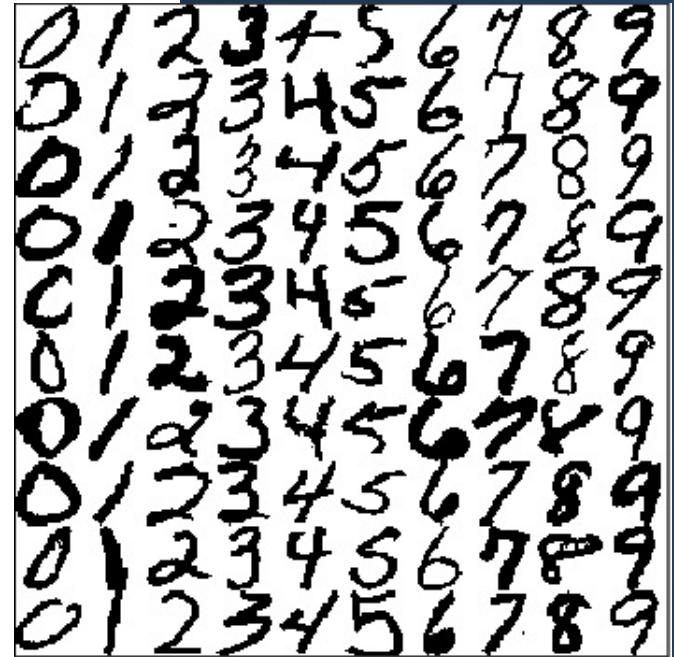
# What is machine learning

Pure Math -> Stats/Probs/Comb Math ->  
Machine Learning -> Deep Learning -> AGI



# What is Machine Learning?

- Arthur Samuel, 1959
  - *Machine Learning*: field of study that gives computers the ability to learn without being explicitly programmed (explicit solution).
    - e.g. computer learns to play checkers by playing against itself.
- There are many applications that are difficult to program by hand.
  - Example (MNIST, 1998):
    - handwritten digit recognition
    - 28x28 image  $\rightarrow$  784-dim vector
    - a lot of variations & permutations
    - difficult to identify rules & code by hand
  - ML solution:
    - gather some examples
    - train computer to discover differences automatically



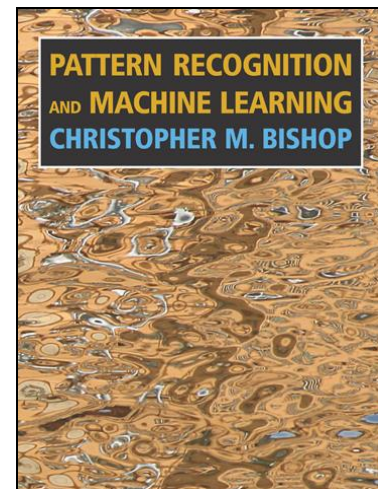
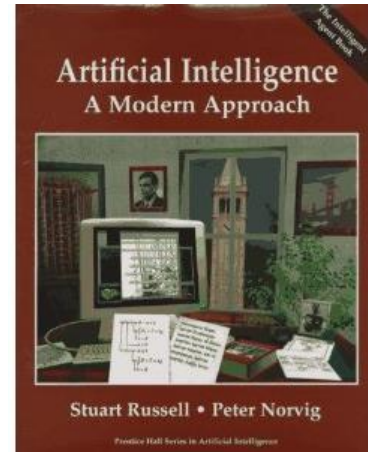
# What is Machine Learning?

- Tom Mitchell, 1997
  - *Well-posed Learning Problem*: "A computer program is said to learn from experience  $E$  with respect to some class of tasks  $T$  and performance measure  $P$ , if its performance at tasks in  $T$ , as measured by  $P$ , improves with experience  $E$ ."
    - e.g., the computer gets better at recognizing digits as it sees more examples, as measured by the error rate.
- A closer look...
  - "*class of tasks  $T$* " (*MNIST classification*)
    - learning is task-specific (recognition, grouping/clustering, ...)
  - "*performance measure  $P$* " (*Classification Percentage*)
    - optimize a loss function (e.g., error rate), but also prevent overfitting (regularization). "generalization"
  - "*experience  $E$* " (*Dataset!*)
    - data-driven! more data is better!



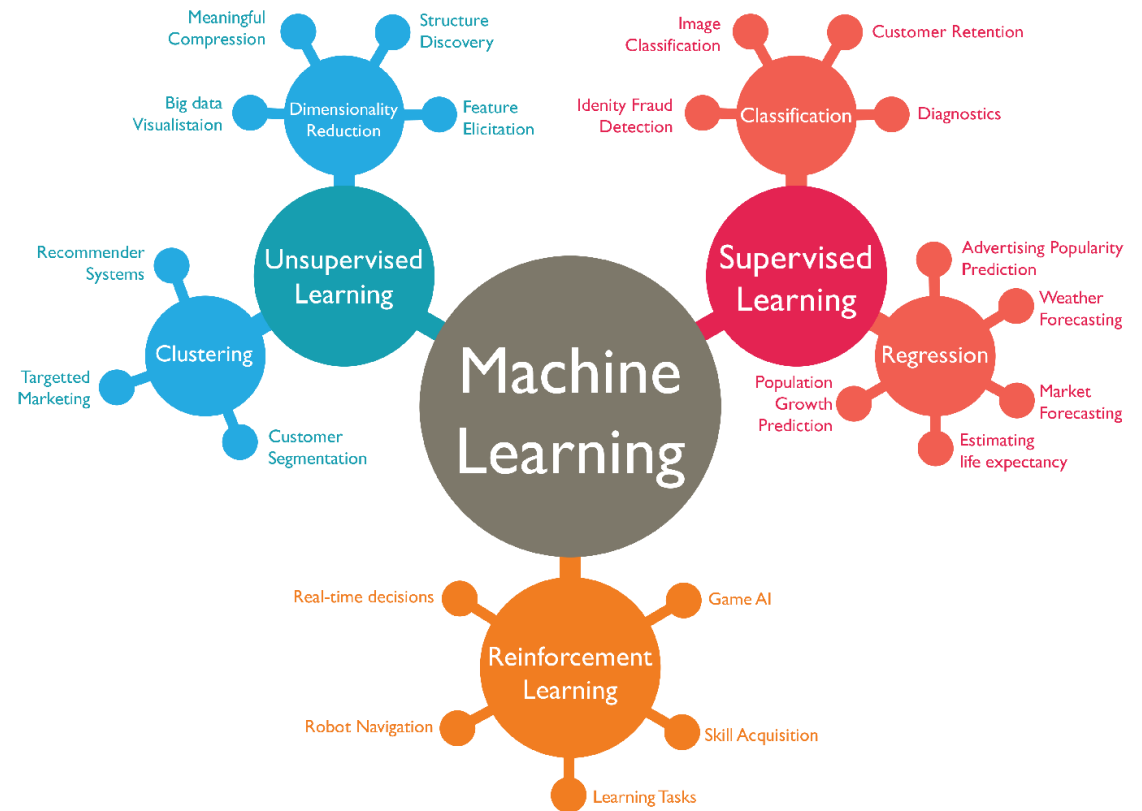
# ML vs AI

- Machine learning grew out of early work in AI
  - and other fields: statistics, physics, neuroscience, ...
  - fueled by more powerful computers and more data.
- "Traditional" Artificial Intelligence (Russell-Norvig)
  - Turing test (is it a computer or a human?)
  - solving by searching ( $A^*$ ,  $\alpha$ - $\beta$  pruning, game playing)
  - knowledge-based (representation, reasoning, logic)
  - planning, scheduling
  - natural language processing
- Machine Learning (Bishop)
  - probability, statistics, Bayesian formulation
  - regression, classification
  - neural networks, kernel functions, kernel machines
  - graphical models, approximate inference
  - statistical learning theory
  - deep learning (deep neural networks)



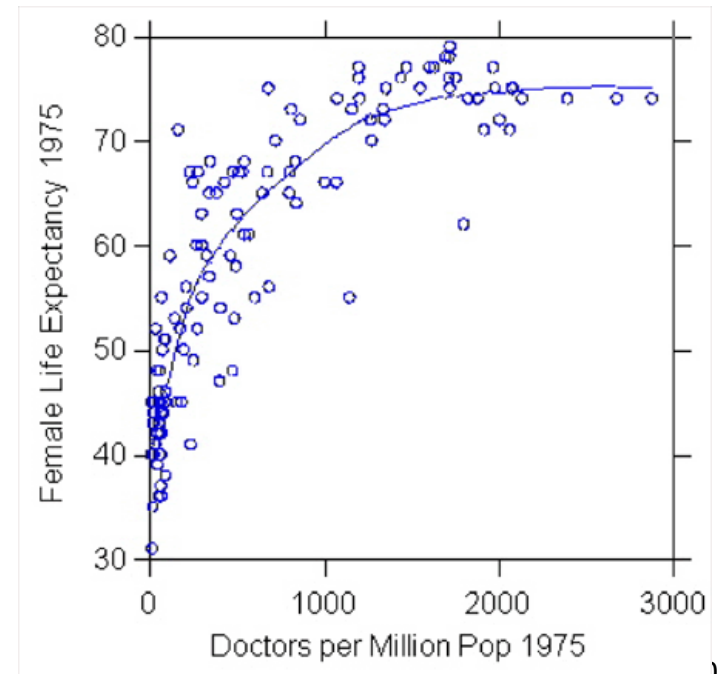
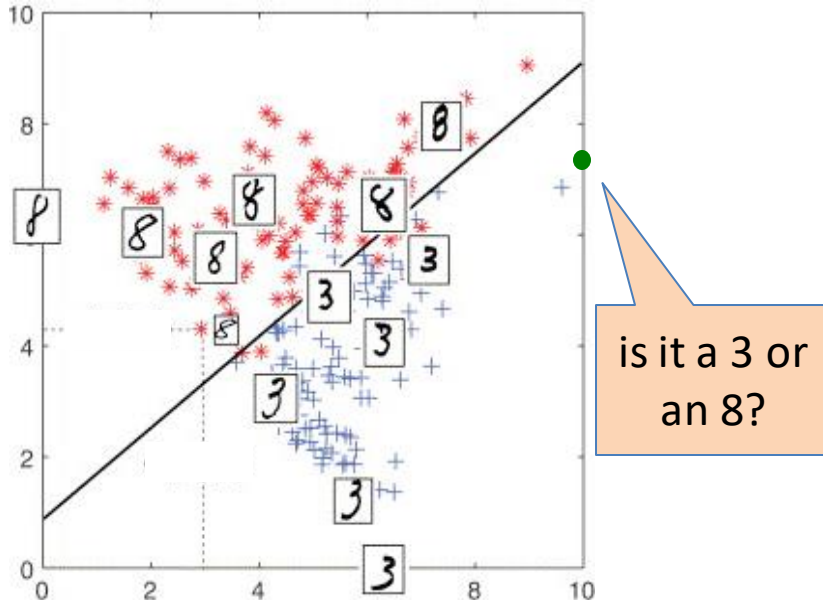


# Topics in ML

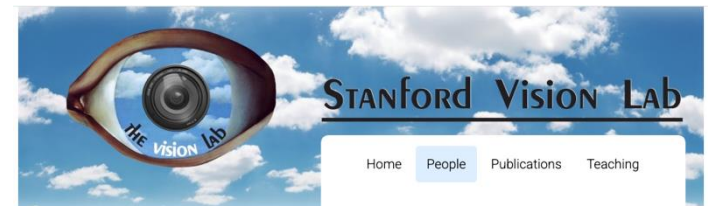


# Topics in Machine Learning

- *Supervised Learning*
  - training data has inputs and outputs
    - Outputs are signal, label, annotation to demonstrate the correct
    - e.g., digit recognition (input=image, output=digit)
  - learn a function mapping **inputs** to **outputs**
    - classification & regression

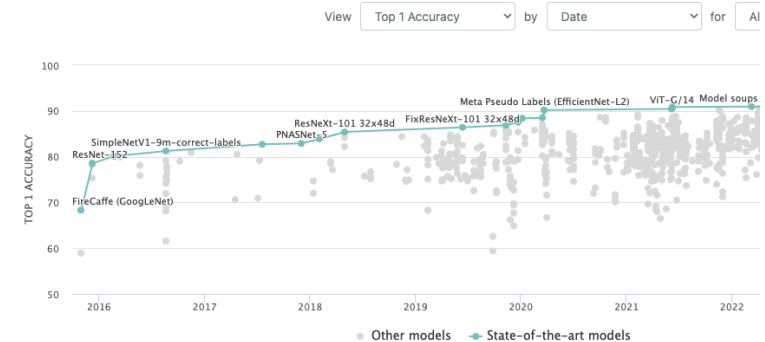


# ImageNet 1000 classes



## Image Classification on ImageNet

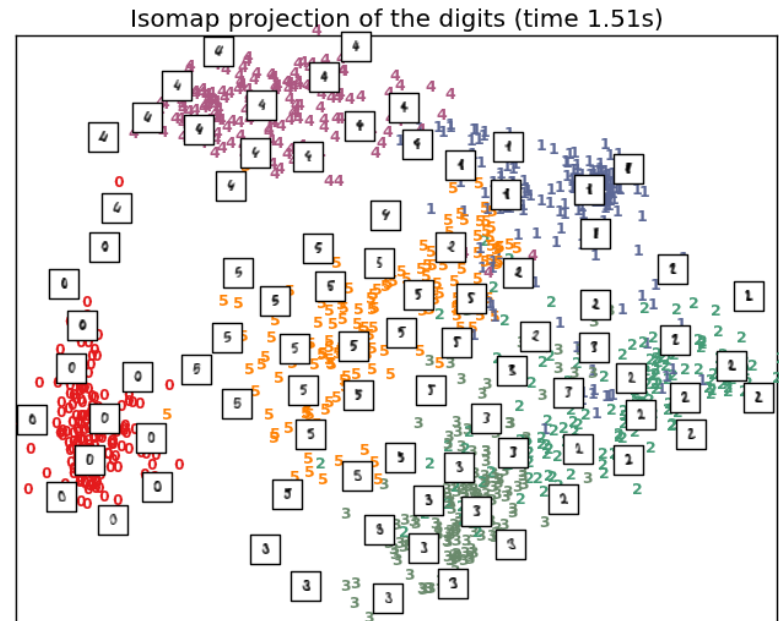
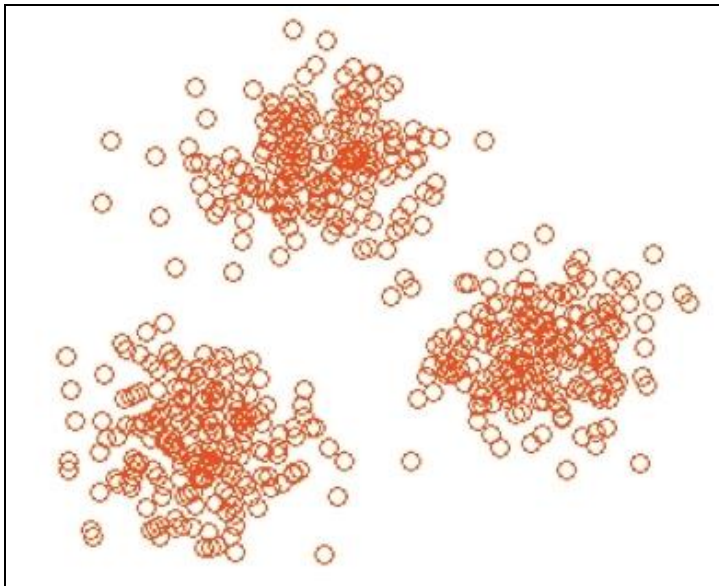
Leaderboard   Dataset



- Neural Architecture Search
- Manual Design
  - ResNet
  - AlexNet
  - VGG
  - ViT
  - Swin Transformer
  - ...

# Topics in Machine Learning

- *Unsupervised Learning (representation learning)*
  - training data only has **inputs** (no outputs as signals)
    - e.g., collection of web documents
  - density estimation - learn a distribution over input space.
  - clustering - discover groups of similar examples.
  - visualization - project high-dim data to 2 or 3-dimensions.



# Self-Supervised learning (before Ilm)

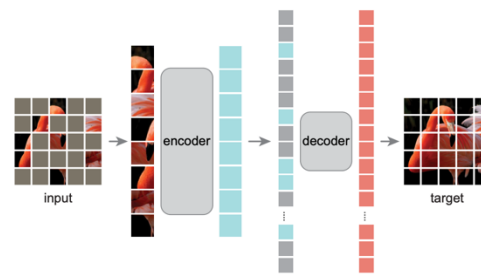


Associate Professor, EECS, MIT

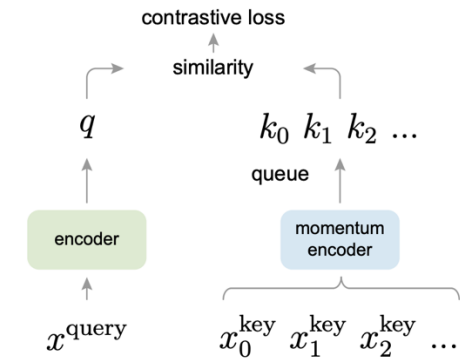
Distinguished Scientist, Google DeepMind

kaiming@mit.edu

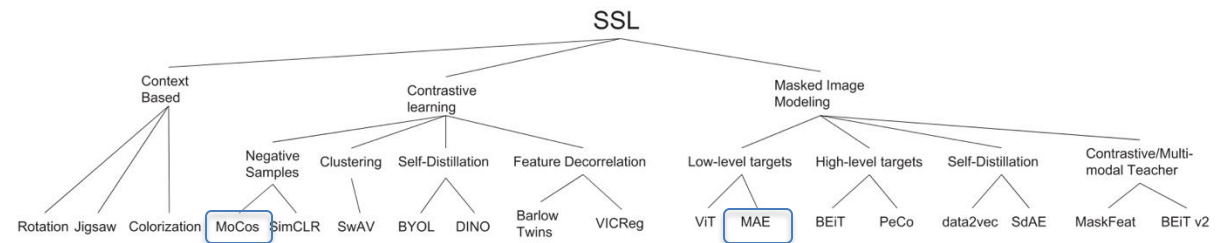
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Masked Autoencoder



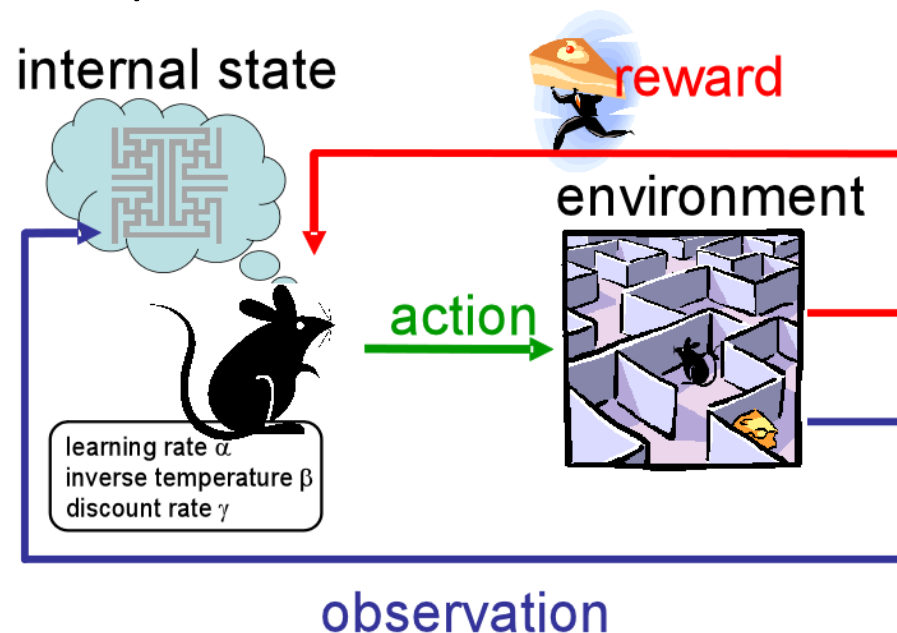
Momentum Contrast





# Topics in Machine Learning

- *Reinforcement Learning (Autonomous Driving, Robotics)*
  - make a sequence of actions, given current states
    - e.g. a robot interacting with its environment
    - Typically as a Markov decision process (MDP)
  - maximize the reward
    - at some point, receive a reward or a punishment.
    - actions may also affect future reward.



# DPO for post-training of LLM/VLM/GenAI models

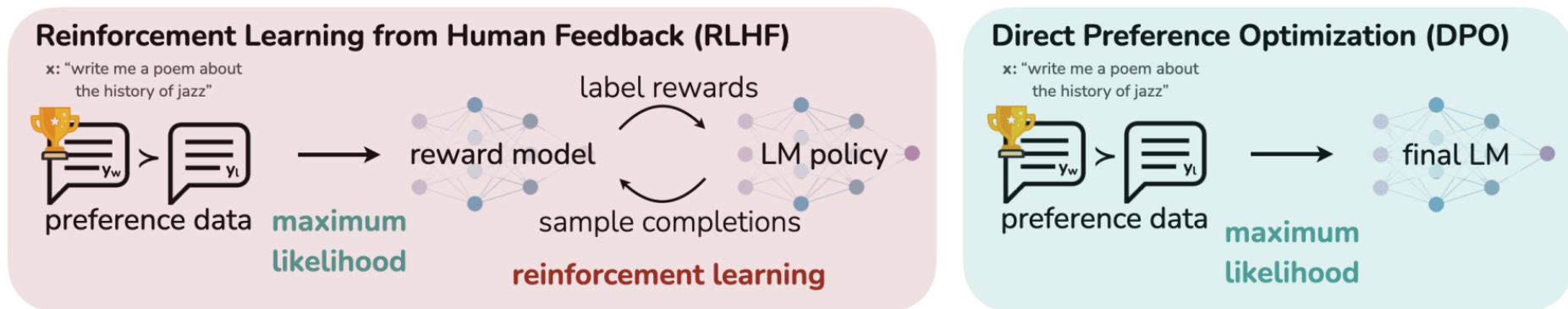
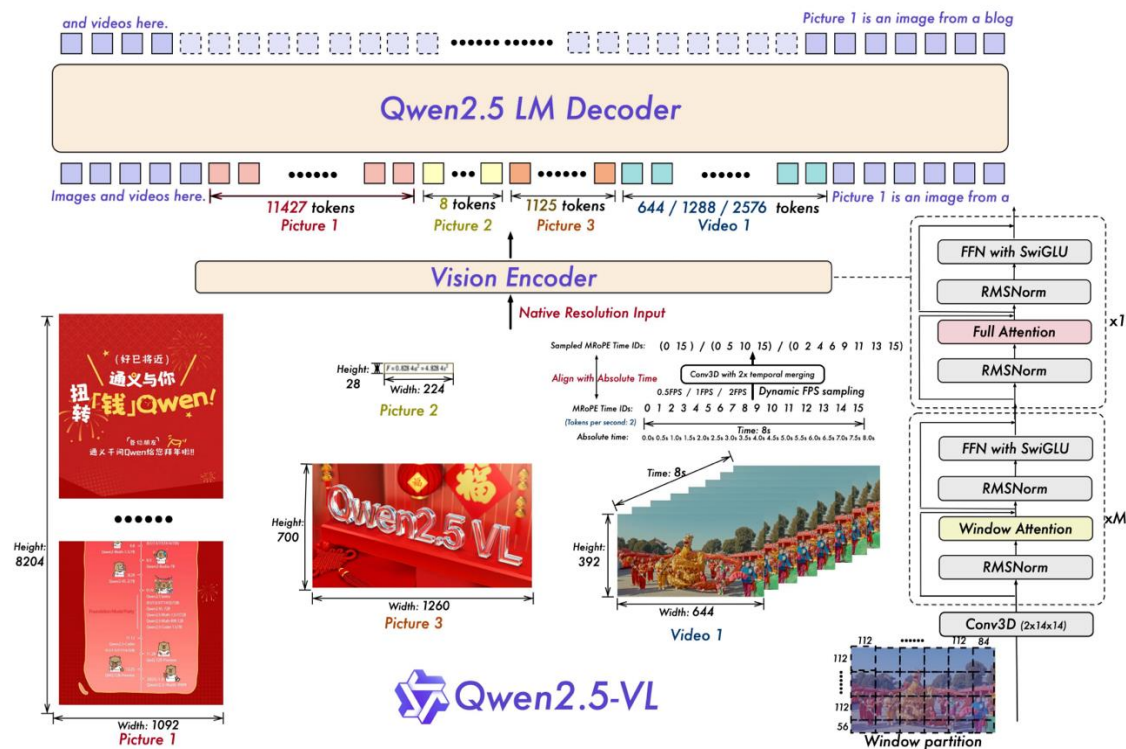


Figure 1: **DPO optimizes for human preferences while avoiding reinforcement learning.** Existing methods for fine-tuning language models with human feedback first fit a reward model to a dataset of prompts and human preferences over pairs of responses, and then use RL to find a policy that maximizes the learned reward. In contrast, DPO directly optimizes for the policy best satisfying the preferences with a simple classification objective, fitting an *implicit* reward model whose corresponding optimal policy can be extracted in closed form.



The best VLM currently. You may already use it somewhere.

How are  
LLMs/VLMs  
trained?



# Single choice

- A. Supervised learning
- B. Unsupervised learning
- C. Reinforcement learning
- D. Mixture of the above 3

# Mixture

## 3 Pre-training

## Self-supervised

Our language model pre-training process consists of several key components. First, we carefully curate high-quality training data through sophisticated filtering and scoring mechanisms, combined with strategic data mixture. Second, we conduct extensive research on hyperparameter optimization to effectively train models at various scales. Finally, we incorporate specialized long-context pre-training to enhance the model's ability to process and understand extended sequences. Below, we detail our approaches to data preparation, hyperparameter selection, and long-context training.

## 4 Post-training

Qwen 2.5 introduces two significant advancements in its post-training design compared to Qwen 2:

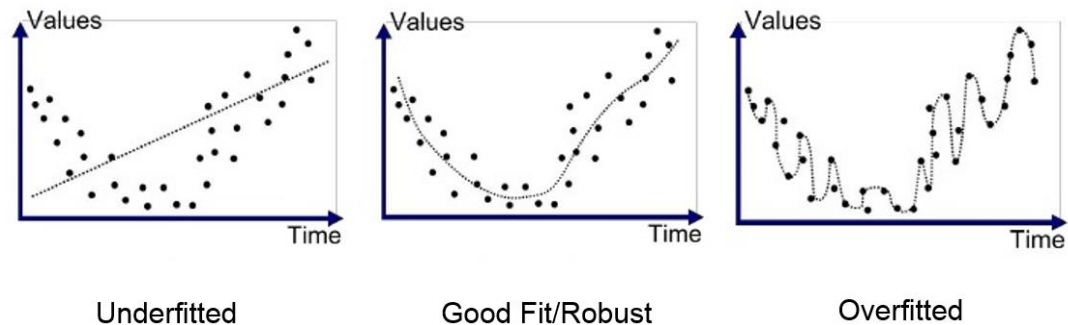
- (1) **Expanded Supervised Fine-tuning Data Coverage:** The supervised fine-tuning process leverages a massive dataset comprising millions of high-quality examples. This expansion specifically addresses key areas where the previous model showed limitations, such as long-sequence generation, mathematical problem-solving, coding, instruction-following, structured data understanding, logical reasoning, cross-lingual transfer, and robust system instruction.
- (2) **Two-stage Reinforcement Learning:** The reinforcement learning (RL) process in Qwen 2.5 is divided into two distinct stages: Offline RL and Online RL.
  - *Offline RL:* This stage focuses on developing capabilities that are challenging for the reward model to evaluate, such as reasoning, factuality, and instruction-following. Through meticulous construction and validation of training data, we ensure that the Offline RL signals are both learnable and reliable (Xiang et al., 2024), enabling the model to acquire those complex skills effectively.
  - *Online RL:* The Online RL phase leverages the reward model's ability to detect nuances in output quality, including truthfulness, helpfulness, conciseness, relevance, harmlessness and debiasing. It enables the model to generate responses that are precise, coherent, and well-structured while maintaining safety and readability. As a result, the model's outputs consistently meet human quality standards and expectations.

## Supervised

## Reinforcement Learning

# Topics in Machine Learning

- *Learning Theory (overfitting or not?)*
  - why does machine learning work? (Generalizability)
    - performance guarantees – bounds on the expected test error.
  - what types of functions can be represented by an algorithm, and how much data do we need?
    - empirical risk minimization, VC dimension, PAC learning

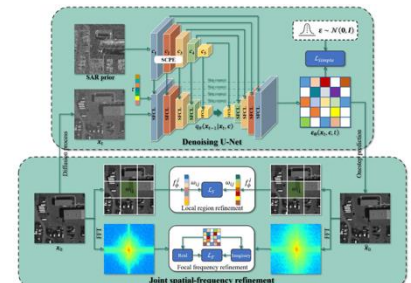
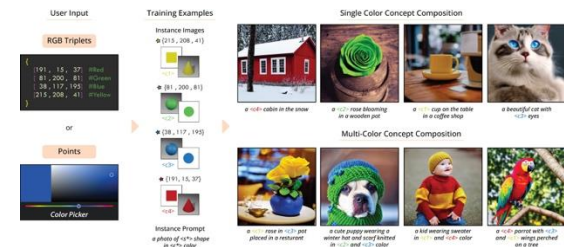
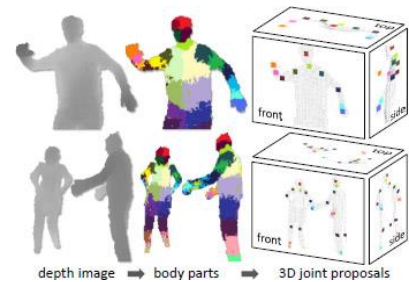


Nowadays with large-scale data, LLM should overfit or not?

- Yes
- No

# ML in the Real World

- Google
  - DeepMind
  - spam email classifier, speech recognition, **AlphaGo**, **AlphaFold** machine translation, image search, quick links,
- Face detection & recognition
  - digital cameras, Google street view, Facebook
- Business
  - credit card fraud detection
  - stock trading (portfolio optimization)
- Recommendation systems
  - Netflix, Amazon
- Human pose recognition (Kinect)
- Controllers (reinforcement learning)
- **Self-driving Cars**
  - **Tesla**
- **AIGC (ChatGPT, Dall-E, Qwen, Hunyuan)**
  - What I'm working on



# Building Blocks for ML

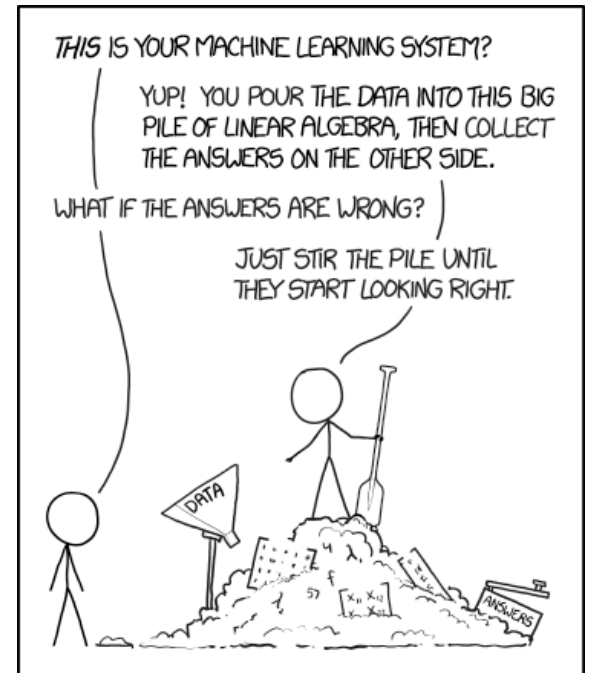
- What tools do we need?
  - *linear algebra*
    - matrices, inverse, eigenvector, SVD, ...
  - *calculus*
    - integrals, derivatives, ...
  - *probability & statistics*
    - random variables, expectation, Bayes' theorem, ...
  - *information theory*
    - entropy, KL divergence, mutual information
  - *optimization theory*
    - Lagrange multipliers, duality, KKT condition

(don't worry, we will review these concepts when necessary...)

(also, more resources are on the course website)



- <https://xkcd.com/1838/>



# Course Intended Learning Outcomes (CILOs)

1. Identify and explain common machine learning algorithms;
2. Implement machine learning algorithms and apply them to solve real-world problems;
3. Analyze and evaluate the effectiveness of different machine learning algorithms, and assess their relative merits;
4. Design and create new machine learning algorithms to address algorithmic shortcomings and solve particular problems.

# Teaching and Learning Activities (TLAs)

- *Lecture* (2 hrs)
  - lecture on the whiteboard (iPad)...there are **no** ppt slides!
  - *Why?*
    - slows down the pace; see the material 3 times.
    - think critically (e.g., teach *how* to derive)
    - active vs. passive learning
  - *What if I make a mistake copying down my notes?*
    - part of being a postgrad student is being able to catch your own mistakes (being self-critical).
    - check reference textbooks, or discuss with classmates.
- *Course Packet*
  - problems on analyzing and deriving ML algorithms.
  - (also a lot of useful properties/models that are not covered in lecture)
- *Tutorial* (1 hr)
  - work on selected problems from the course packet.

# Teaching and Learning Activities (TLAs)

- *Assignments*
  - theory: selected problems from the course packet.
    - *Try to do these by yourself without reference to other materials.*
    - Are you really learning anything if you are just copying the solutions from the Internet?
  - programming: implementing ML algorithms and try on small datasets.
- *Course Project*
  - use machine learning to solve a real-world problem.
    - e.g., something related to your research.
    - if you don't have a research topic, there will be a recommended course project (e.g., digit recognition).
  - course report and poster presentation (or spotlight videos).
  - **groups of 2.**

# ICAP Theory of Cognitive Engagement

- *Which type of engagement is best for learning?*
  - **Interactive > Constructive > Active > Passive**

|                 | Interactive                                                     | Constructive                                                                                       | Active                                                                                         | Passive          |
|-----------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------|
| Behavior        | Collaborate                                                     | Generate                                                                                           | Manipulate                                                                                     | Attend           |
| Description     | collaborating with a peer, building knowledge co-constructively | producing additional small pieces of knowledge beyond what was presented in instructional material | manipulating instruction material (taking notes, highlighting), but not adding new information | paying attention |
| CS5487 activity | Course project                                                  | Course packet (assignments)                                                                        | Lecture, tutorial                                                                              |                  |

- **Paper:** M. Chi, et al. [“Translating the ICAP Theory of Cognitive Engagement Into Practice”](#), *Cognitive Science*, 2018.
- **Talk:** <https://youtu.be/JLpZJCBj2qY>

# Assessment

- Assignments (30%)
  - *theory* (10%) – 1-2 problems per week
  - *programming* (20%) – 2 programming assignments
- Midterm (10%)
  - you get an A4 cheat sheet (one side, *handwritten by pen or pencil*)
- Course Project (30%)
- Final Exam (30%)
  - 2-hour final examination
  - **exam problems are selected from the course packet.**
  - you get an A4 cheat sheet (both sides, *handwritten by pen or pencil*)
- Requirement:
  - Need at least 30% on the exam and 30% on the project to pass the course.
    - A necessary condition, not a sufficient condition.



# Programming

- For the programming assignments (PA)
  - *You should implement the algorithms using your own code.*
    - Do **NOT** use someone else's implementation (including libraries)
    - You can use common library components and functions (e.g., matrix operations, optimization toolboxes, etc).
    - If you use anything special, make sure to mention it in your report.
    - Part of this course is about how to implement ML algorithms.
  - If you **DO** use someone else's implementation, you must mention it in your report.
    - 10-20% of the marks for the programming assignment are allocated for implementation. If you use a 3<sup>rd</sup> party implementation, you will not get these marks.
- For the course projects
  - You may use 3<sup>rd</sup> party libraries. You must acknowledge the libraries that you use in your project report.

# Sitting in ...

- *Can I just attend the lectures without registering?*
  - The answer is **no**.
- Here are the reasons:
  1. The assignments are also teaching activities that give you hands-on experience (both in theory and programming).
    - If you are going to do these, you might as well do it for credit.
  2. Free to do whatever you want for the course project – apply ML to your research problem / ideas.
    - A good course project could lead to a **publication**.
  3. Machine Learning is an important topic in CS and increasingly important in other fields.
    - It looks good on your transcript/CV.
    - InterView (**CS229+LeetCode**)
- *I can offer exceptions to undergraduate students on a case-by-case basis.*

# CS5487 Schedule (2025A)

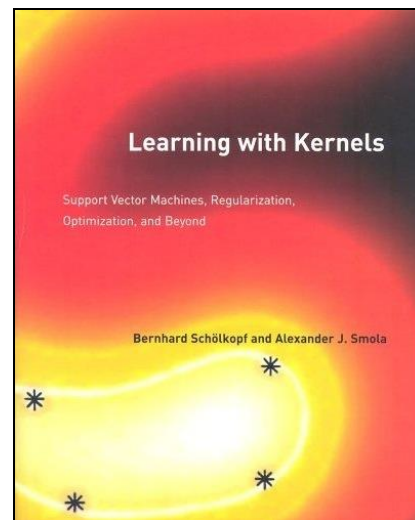
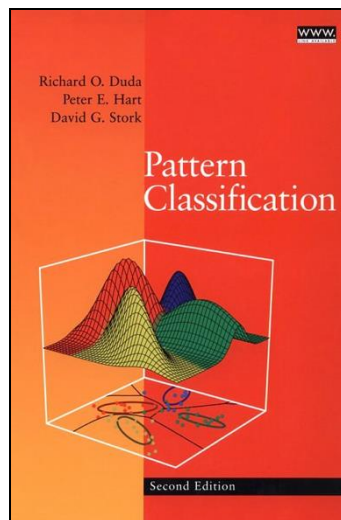
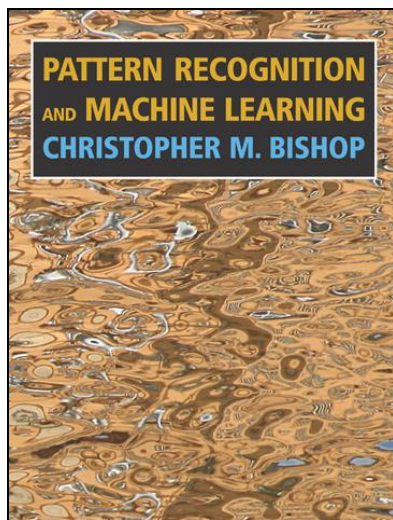
| Wk (M/DD)   | Lecture                                              | Topics                                                                                     | Reference                        | Assessment           |
|-------------|------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------|----------------------|
| 1. (9/02)   | 1. Introduction & Review                             | Probability & Statistics, Linear Algebra                                                   | PRML 1.2, 2, Appendix C          |                      |
| 2. (9/09)   | 2. Parameter Estimation                              | MLE, bias/variance, LS regression                                                          | PRML 2.1-2.4, 3.1; DHS 3.2       |                      |
| 3. (9/16)   | 3. Bayesian Estimation                               | MAP, Bayes estimation, Gaussians                                                           | PRML 2.3, 3.1, 3.3; DHS 3.3-3.5  | PA1 out              |
| 4. (9/23)   | 4. Parametric Clustering                             | K-means, GMM & EM                                                                          | PRML 9; DHS 10.1-10.4            |                      |
| 5. (9/30)   | 5. Non-parametric Clustering                         | KDE & mean-shift                                                                           | PRML 2.5; DHS 4.1-4.4; m-s paper |                      |
| 6. (10/07)  | 6. Bayesian classifiers                              | Bayesian Decision Theory, Gaussian classifier, Naïve Bayes                                 | PRML 1.5; DHS 2.1-2.6            | PA1 due, PA2 out     |
| 7. (10/14)  | <i>Midterm Quiz</i>                                  |                                                                                            |                                  |                      |
| 8. (10/21)  | 7. Dimensionality & Linear Dimensionality Reduction  | PCA, LSA, pPCA, FA, CCA                                                                    | PRML 12-12.2, 12.4; DHS 3.7-3.8  |                      |
| 9. (10/28)  | 8. Discriminative Learning – Linear Classifiers (I)  | least-squares classification, perceptron, logistic regression, empirical risk minimization | PRML 4.1-4.3                     | PA2 due, Project out |
| 10. (11/04) | 9. Discriminative Learning – Linear Classifiers (II) | linear SVM, regularized risk                                                               | PRML 7.1; DHS 5.11               | Project proposal due |
| 11. (11/11) | 10. Kernels                                          | kernel functions, kernel SVM                                                               | PRML 6.1-6.2, 6.4, 7.1           |                      |
| 12. (11/18) | 11. Non-linear Dim. Reduction                        | kernel PCA, pre-image problem                                                              | PRML 12.3                        |                      |
| 13. (11/25) | 12. Bonus lecture                                    | TBA                                                                                        |                                  |                      |
| 14. (12/06) |                                                      | Course Project Presentations                                                               |                                  | Project report due   |

# About Due Dates

- Note: all due dates are "soft", in the sense that I won't take points off if you are late during the semester.
- On the other hand, you should try to do the assigned HW problems week-by-week, as this will help you understand the concepts after you have seen them in class.
- The programming assignment should also be done in a reasonable time frame, since they will give you some hands-on experience using the algorithms, and may give you some ideas for your course project later.
- The **hard** deadline for **all** course work (problem sets, programming assignments, course project) is **Week 14, Friday, Dec 5th @ 5pm**.
  - I will handout solutions to the problem sets *after* this deadline.
  - ***Any coursework submitted after this deadline will not be graded.***

# Learning Resources

- Course website on Canvas – CS5487
  - announcements, course packet, discussion board
  - To practice is better than only to read
- Textbooks
  - **Bishop, *Pattern Recognition and Machine Learning*.**
  - Duda, Hart, & Stork, *Pattern Classification*.
  - Schölkopf and Smola, *Learning with kernels: support vector machines, regularization, optimization, and Beyond*.



# Important Publication Venues

- *ML Conferences*
  - Neural Information Processing Systems (NeurIPS)
  - International Conf. on Machine Learning (ICML)
  - International Conf. on Learning Representations (ICLR)
- *CV Conferences*
  - CVPR
  - ECCV
  - ICCV

# Computer Resources

- Programming Languages for Machine Learning
  - **Python (numpy, scipy, sklearn; installed in CS lab)**
    - CS Jupyter hub server; use EID to login.
      - <https://ltjh.cs.cityu.edu.hk/>
  - MATLAB (university site license)
  - Octave (open-source clone of MATLAB)
  - C++ (OpenCV library)
- Tutorials
  - Python:
    - <https://docs.scipy.org/doc/numpy/user/quickstart.html>
    - <http://scikit-learn.org/stable/tutorial/index.html>
  - Matlab:
    - [http://www.mathworks.com/academia/student\\_center/tutorials/launchpad.html](http://www.mathworks.com/academia/student_center/tutorials/launchpad.html)
    - [http://www.stanford.edu/~wfsharpe/mia/mat/mia\\_mat3.htm](http://www.stanford.edu/~wfsharpe/mia/mat/mia_mat3.htm)
    - <http://www.math.mtu.edu/~msgocken/intro/intro.html>
  - Octave: <http://www.gnu.org/software/octave/>

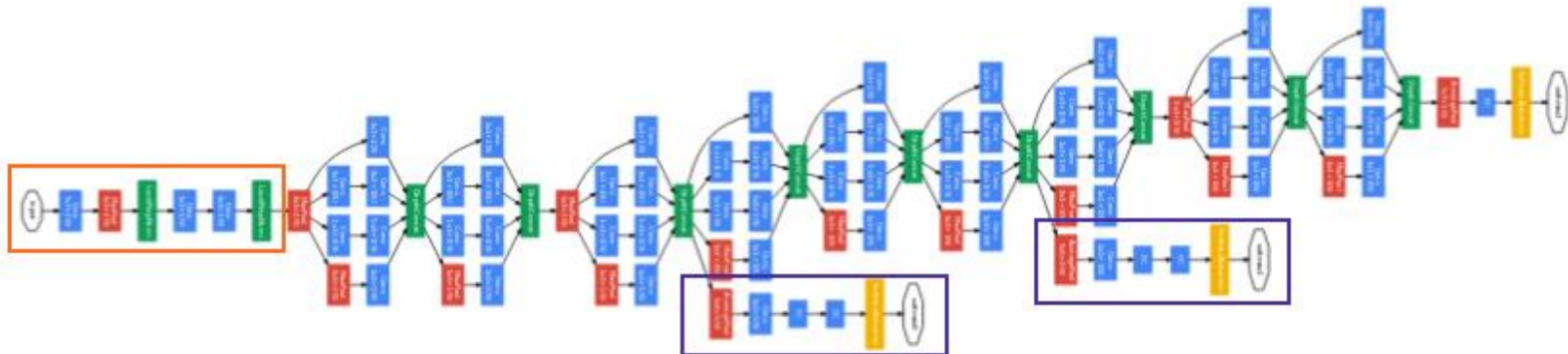
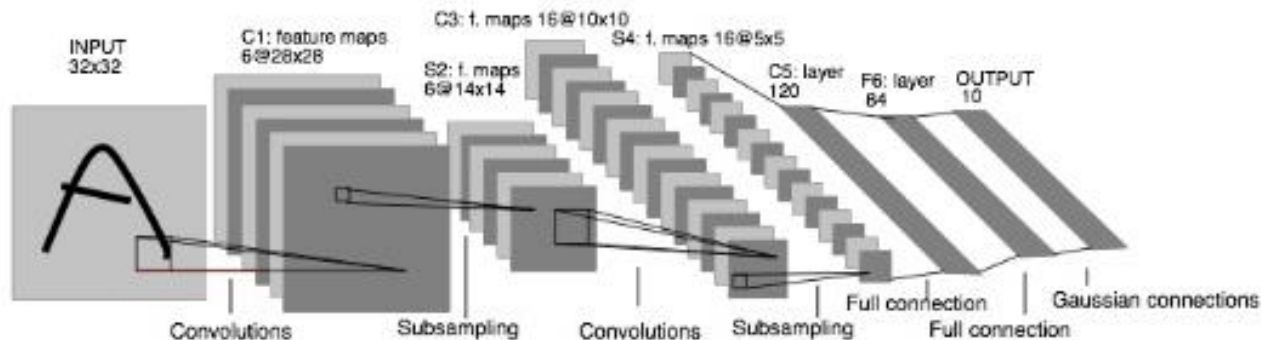


# Academic Honesty

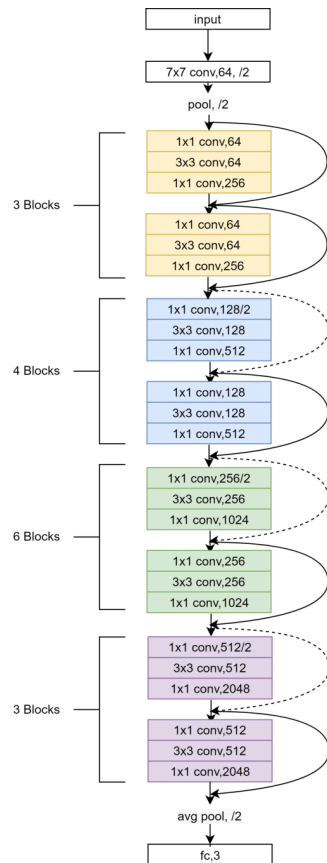
- CityU has *Rules of Academic Honesty* and has required all students to complete an online tutorial on subject and declare your understanding
  - Plagiarism...
    - It is serious fraud to plagiarize others' work.
    - Punishment ranges from warning to course failure.
  - How to prevent plagiarism...
    - Finish the assignments by yourself! You have to write the program/solution yourself.
      - okay to talk about *how* to do the problem with your classmates.
      - **Protect your code**; don't give it away as a "reference" copy.
      - **Using AI! Different prompts!**
    - In plagiarism cases, we treat the giver and the copier as both guilty.
    - You hurt your own grades by not reporting cheating.
  - As instructors...
    - We have responsibility to report academic dishonesty cases so as not to compromise the quality of education
    - We take suspected plagiarism cases very seriously.
- CS5487— All plagiarism cases are sent to the College of Engineering.

# Neural Networks & Deep Learning

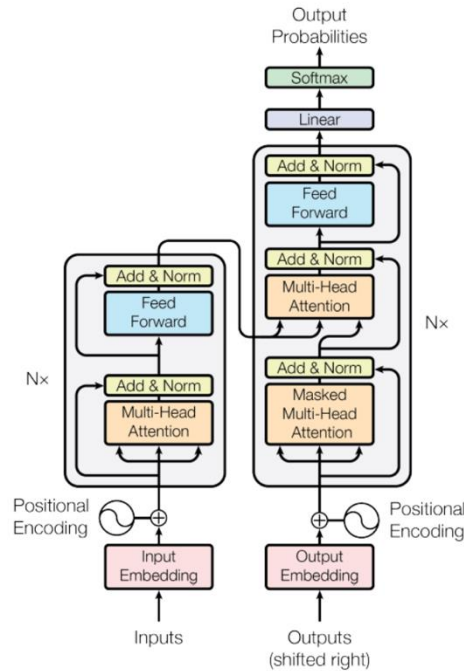
- *Deep learning* is supervised/unsupervised learning using multi-layer neural networks.
  - Improved training algorithms to prevent overfitting
  - Faster parallelization (GPUs)
  - More data (millions of examples)



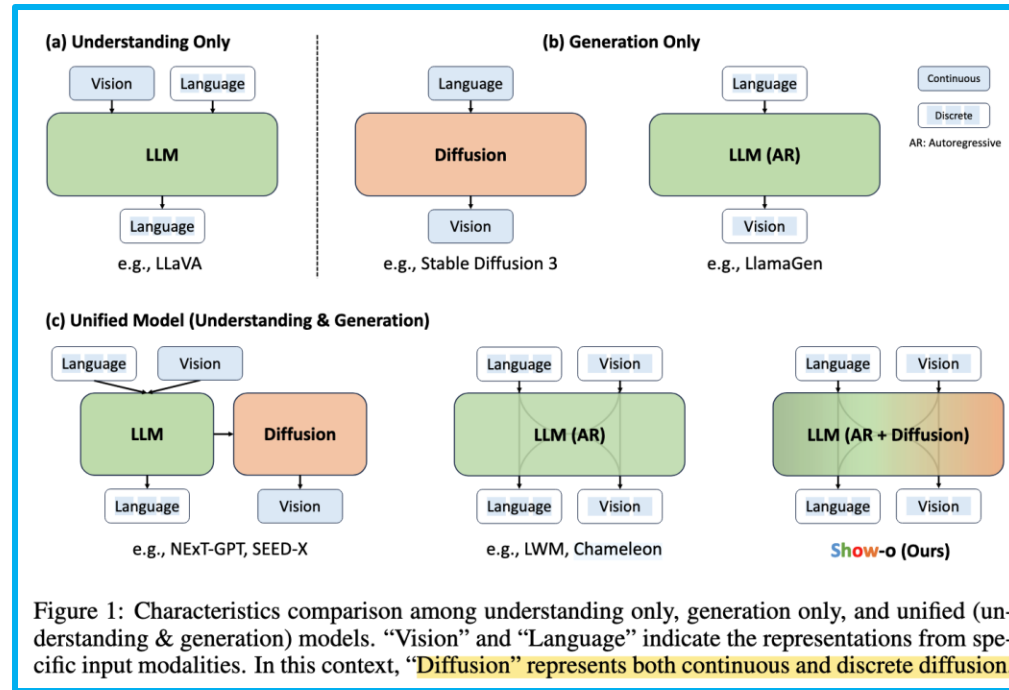
# Current popular architectures



ResNet  
(Kaiming He)  
CS5487

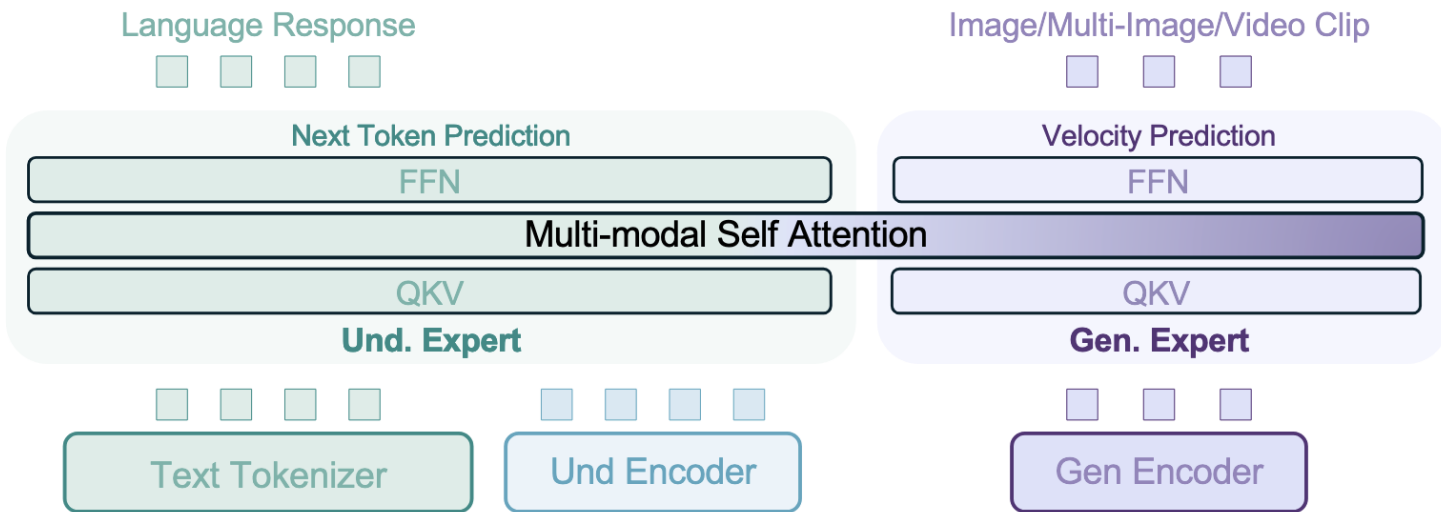


Transformer



Mixed architecture mainly with ViT

# BAGEL model from Bytedance



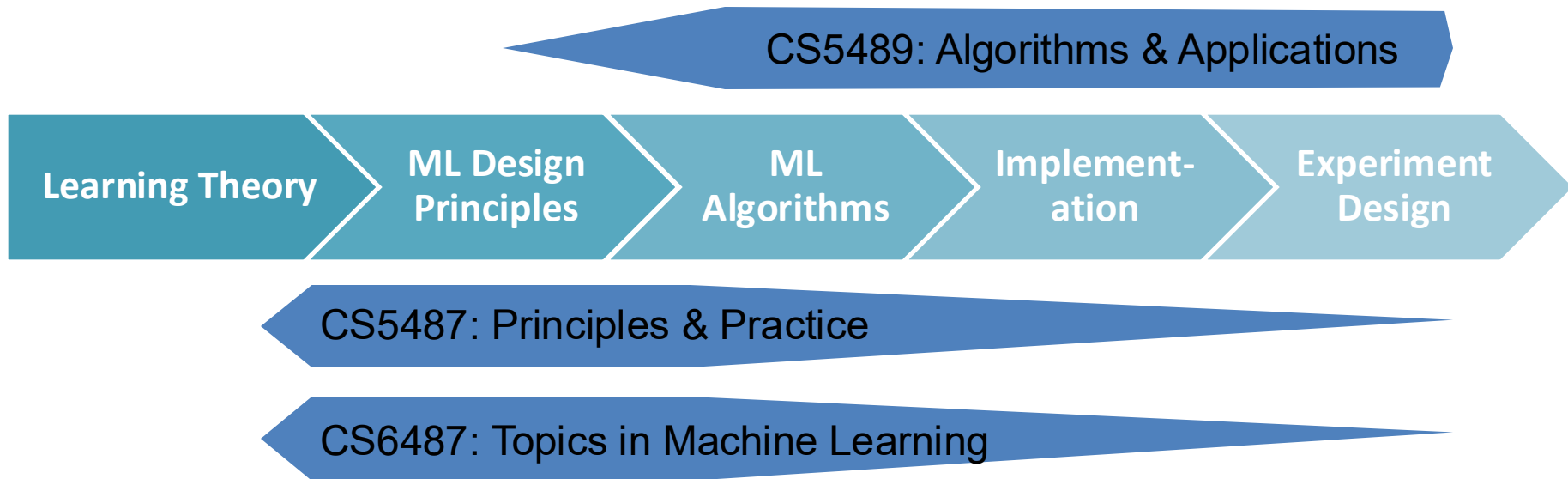
**Figure 2** We use **two Transformer experts** to process understanding and generation information, and all tokens do shared multi-modal self attention in each Transformer block. We adopt two distinct encoders to separately capture semantic content and low-level pixel information for image understanding and generation tasks.

# Related Courses

- CS5489 Machine Learning: Algorithms & Applications
  - Prof. Kede Ma
  - Topics: machine learning applications, deep learning, experiment design.
- CS6487 Topics in Machine Learning
  - Topics: varies by lecturer
- CS5486 Intelligent Systems
  - Topics: knowledge systems, fuzzy systems, rule-based systems, evolutionary computation.
- CS5491 Artificial Intelligence
  - Topics: Knowledge representation, uncertainty reasoning, reinforcement learning, AI-based search.
- CS6493 Natural Language Processing
  - Topics: computational linguistics, statistical and neural-based language models.

# Relationship with other ML courses

- CS5489 mainly focuses on the intuition of how machine learning algorithms work, implementation of algorithms, applying machine learning to applications, and analyzing the results.



- CS5487/CS6487 mainly focus on ML design principles and derivation of algorithms.

# Machine Learning



what society thinks I do

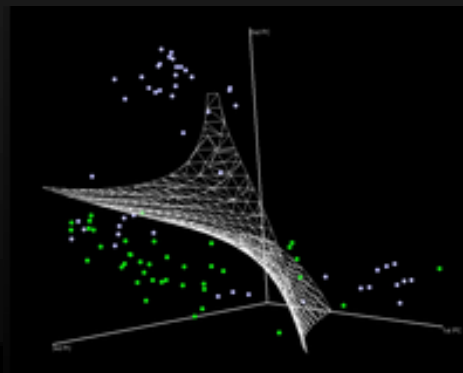


what my friends think I do



what my parents think I do

$$\begin{aligned}
 L_p &= \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^n \alpha_i y_i (\mathbf{x}_i \cdot \mathbf{w} + b) + \sum_{i=1}^n \alpha_i \\
 \alpha_i &\geq 0, \forall i \\
 \mathbf{w} &= \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i, \sum_{i=1}^n \alpha_i y_i = 0 \\
 \nabla \hat{g}(\theta_t) &= \frac{1}{n} \sum_{i=1}^n \nabla \ell(x_i, y_i; \theta_t) + \nabla r(\theta_t). \\
 \theta_{t+1} &= \theta_t - \eta_t \nabla \ell(x_{i(t)}, y_{i(t)}; \theta_t) - \eta_t \cdot \nabla r(\theta_t) \\
 \mathbb{E}_{i(t)}[\ell(x_{i(t)}, y_{i(t)}; \theta_t)] &= \frac{1}{n} \sum_i \ell(x_i, y_i; \theta_t).
 \end{aligned}$$



```
>>> from scipy import SVM
```

what other programmers think I do

what I think I do

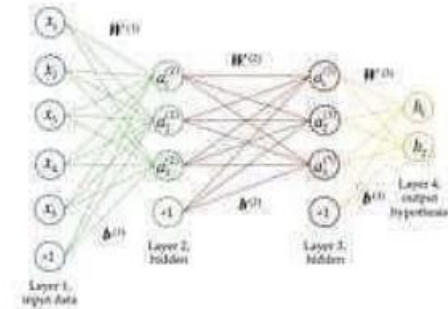
what I really do



A parrot



Machine learning algorithm



Learns random phrases



Doesn't understand anything about what it learns



Occasionally speaks nonsense



Is a cute birdie parrot



# Probability and Statistics

- Random variable
- Event
- Discrete/Continuous examples
  - Discrete: Bernoulli / Binomial / Poisson distributions
  - Continuous: Gaussian Distribution
- Central limit theorem
- Joint/marginal/conditional distribution
- Independence
- Bayes Rule
- Expectation/Mean/Variance/Covariance

# Linear Algebra

- Vector
- Matrix
- Properties
- Operations
- **Tutorials!**